# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 2 | **(d)** 36 g, 1 : 8 | 4 + 32 = 36; 4:32 = 1:8. |
| 3 | **(c)** homogeneous mixture | Alloy. |
| 4 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 5 | **(d)** 1 g and 8 g | Ratio 1 : 8. |
| 6 | **(d)** hydrogen gas | Metal with acid. |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 8 | **(a)** 14 | 2 x 7. |
| 9 | **(a)** silver | Silver tarnish. |
| 10 | **(b)** Both A and R are true but R is not the correct explanation of A. | Both are true but R only restates A as an exception. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Oxygen = 36 - 4 = 32 g (1); Ratio = 4 : 32 = 1 : 8 (1). | 1+1 |
| 12 | They have a lower melting point than the metals (1); Allow easy joining by melting (1). | 1+1 |
| 13 | Their outer arrangement is stable (1); Used where reaction is unwanted, as in bulbs (1). | 1+1 |
| 14 | One molecule has 3 atoms (1); 3 x 3 = 9 atoms (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Probably iron, an element (1); Magnetic and reacts with acid giving hydrogen (1); Could also be a mixture containing iron (1). | 1+1+1 |
| 16 | In diamond all electrons are held in fixed bonds (1); In graphite some electrons are free to move (1); Different arrangements (1). | 1+1+1 |
| 17 | A homogeneous mixture (1); Evaporate and condense water (1); Distillation (1). | 1+1+1 |
| | **OR:** Sodium: soft reactive metal (1); Chlorine: poisonous greenish gas (1); NaCl: harmless white crystalline solid (1). | 1+1+1 |
| 18 | All dissolve completely (1); Uniform composition (1); Components cannot be seen separately (1). | 1+1+1 |
| | **OR:** Heat sugar in a dry tube (1); Black residue forms (1); Carbon is present (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Prepare a mixture of iron and sulphur (1); Test with a magnet before heating (1); Heat strongly in a test tube (1); Black, non-magnetic product forms after heating (1); A compound is formed; mixtures retain properties (1). | 1+1+1+1+1 |
| | **OR:** Oxygen: molecules of two oxygen atoms (1); Water: molecules with H and O atoms bonded (1); Air: different molecules mixed (1); Atoms joined in compounds (1); Mixtures are not bonded (1). | 1+1+1+1+1 |
| 20 | Compound has a fixed mass ratio (1); Hydrogen = 45 / 9 = 5 g (1); Oxygen = 40 g (1); Same ratio whatever the source (1); Mixtures do not obey this (1). | 1+1+1+1+1 |
| | **OR:** Metal oxides are basic (1); Non-metal oxides are acidic (1); Metals react with acids to give hydrogen (1); Physical: lustre, ductility, sonority (1); Exceptions need care (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** $3O_2$ | Coefficient before formula. |
| 2 | **(c)** sulphur | SO2. |
| 3 | **(a)** Both A and R are true and R is the correct explanation of A. | R is a supporting reason. |
| 4 | **(c)** a compound copper oxide formed | Copper reacts with oxygen. |
| 5 | **(a)** shows properties of both metals and non-metals | Intermediate. |
| 6 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 7 | **(c)** a metal | Metallic properties. |
| 8 | **(d)** semiconductor | Controlled conduction. |
| 9 | **(c)** A is true but R is false. | A is true; R is false. |
| 10 | **(d)** basic in nature | Metal oxides are basic. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | It does not rust (1); It is strong and easy to clean (1). | 1+1 |
| 12 | Distilled water is pure; same throughout (1); Milk is a mixture of several substances (1). | 1+1 |
| 13 | In a mixture each substance keeps its properties (1); The compound is a new black substance (1). | 1+1 |
| 14 | It is less dense and reactive (1); Produces hydrogen and heat (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | H2O: 2 H and 1 O (1); H2O2: 2 H and 2 O (1); Different formulae, different compounds (1). | 1+1+1 |
| 16 | Dazzling white flame (1); White powder forms (1); Magnesium oxide, a compound (1). | 1+1+1 |
| 17 | Highly reactive (1); Combine with oxygen and water (1); Found as compounds (1). | 1+1+1 |
|  | **OR:** Copper = 450 g (1); Tin = 50 g (1); Total 500 g (1). | 1+1+1 |
| 18 | Rusting is chemical and forms iron oxide (1); Dissolving is physical (1); Sugar can be recovered (1). | 1+1+1 |
|  | **OR:** Mg gives basic oxide (1); S gives acidic oxide (1); Tested with litmus (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Metals: conduct, malleable (1); Non-metals: brittle, insulators (1); Metalloids: semiconductors (1); Exceptions: mercury, graphite, bromine (1); Uses: wires, fuels, chips (1). | 1+1+1+1+1 |
|  | **OR:** Sedimentation: physical (1); Filtration: physical (1); Chlorination: chemical (1); Distillation: physical (1); Purpose of each (1). | 1+1+1+1+1 |
| 20 | Magnet to remove iron (1); Filter to remove sand (1); Evaporate to obtain salt (1); Condense vapour for water (1); Physical properties used: magnetism, particle size, boiling point (1). | 1+1+1+1+1 |
|  | **OR:** Elements are also pure substances (1); Compounds are pure with fixed composition (1); Examples: iron, oxygen (elements) (1); Water, salt (compounds) (1); Mixtures are not pure (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** constituents cannot be separated by simple physical methods | Chemical bonds. |
| 2 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 3 | **(d)** blue litmus red | Acidic. |
| 4 | **(d)** a suspension (heterogeneous) | Particles settle. |
| 5 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 6 | **(d)** wood | Non-metal. |
| 7 | **(d)** electrolysis of water | Chemical decomposition. |
| 8 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 9 | **(a)** a salad | Components visible. |
| 10 | **(a)** compound | Breaks into carbon and water. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Non-metal (1); e.g. sulphur gives sulphur dioxide (1). | 1+1 |
| 12 | C: 6, H: 12, O: 6 (1); Total 24 atoms (1). | 1+1 |
| 13 | Metalloids show mixed properties (1); Exceptions like graphite and mercury exist (1). | 1+1 |
| 14 | Iron displaces copper (1); Chemical change (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Brass is an alloy of copper and zinc, unreactive with moisture (1); Iron combines with oxygen and water to rust (1); Alloying changes properties (1). | 1+1+1 |
| 16 | They contain more than one metal (1); Composition varies (1); They are alloys, i.e. mixtures (1). | 1+1+1 |
| 17 | Copper conducts (1); Plastic insulates (1); Prevents shocks (1). | 1+1+1 |
|  | **OR:** Magnet removes iron: magnetic property (1); Heat the rest: ammonium chloride sublimes and is collected (1); Sand remains behind (1). | 1+1+1 |
| 18 | Mercury is a liquid metal (1); Bromine is a liquid non-metal (1); Gallium melts in the hand (1). | 1+1+1 |
|  | **OR:** Contain many components (1); Composition varies (1); Components keep properties (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Stainless steel: iron, chromium, nickel, rust-proof (1); Brass: copper and zinc, strong, golden (1); Solder: tin and lead, low melting point (1); Improved hardness, resistance to corrosion (1); Used in tools, instruments and electronics (1). | 1+1+1+1+1 |
|  | **OR:** Metals: Fe, Cu, Al, Zn, Au (2); Non-metals: C, S, O, N, Cl (2); Conductors: metals (and graphite) (1). | 2+2+1 |
| 20 | Dissolving: physical, mixture (1); Heating iron-sulphur: chemical, compound (1); Rusting: chemical (1); Melting wax: physical (1); Physical changes are reversible, chemical form new substances (1). | 1+1+1+1+1 |
|  | **OR:** Heat the mixture (1); Ammonium chloride sublimes (1); Collect the vapour on a cold surface (1); Sand remains (1); Property: sublimation (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.